

Abstract No.

Dissecting spatial niches and immunological cues in glioblastoma for novel biomarkers and precision therapies

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Introduction: Glioblastoma (GBM) is characterized by its heterogeneity, which renders standardized treatment largely ineffective, underlining the need for novel patient-tailored combination strategies. HOXA9 and WNT6 are oncogenic factors associated with worse GBM prognosis. HOXA9 and WNT6 have been shown to sustain immunoregulatory roles in various contexts, and in GBM we have found an association between these molecules and several immune-related markers and processes. Thus, we aim to explore HOXA9/WNT6-driven effects on GBM-immune spatial architecture, unveiling potential therapeutic strategies.

Materials and Methods: GBM patient samples analyzed by Visium (10X Genomics; Ravi et al., 2022; Tirosh et al., 2024) were used to assess the co-expression of *HOXA9/WNT6* with immune cell types. Additionally, GBM patients from TCGA were stratified into *HOXA9/WNT6*-high and -low groups for differential expression analysis, with selected genes subjected to literature querying and univariate survival analysis. Pharmacologic inhibition of HOXA9 (direct by DB818, indirect by buparlisib) and WNT signaling (E7386) were assessed in four paired HOXA9- and WNT6-modulated cell models.

Results: *HOXA9/WNT6* show spatial co-expression with multiple immune signatures in GBM, particularly of T cells. *HOXA9/WNT6* double-high patients present a downregulation of several genes associated with better survival. In vitro, HOXA9-high cells show greater sensitivity to pharmacologic HOXA9 inhibition. WNT signaling inhibition decreases WNT6-driven aggressiveness in overexpression models and acts synergistically with WNT6 silencing in knockdown models.

Conclusions: These findings shed light on the association between HOXA9/WNT6 and GBM immune landscape and suggest their pharmacologic targeting as a potential therapeutic avenue. Importantly, ongoing spatial transcriptomics and combination treatment studies, using co-cultures with hPBMCs and humanized in vivo models, will further explore this axis, potentially contributing toward more effective GBM care.



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